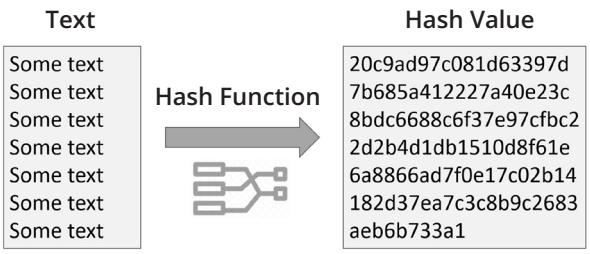
How Hashing is Used in Bitcoin Transactions

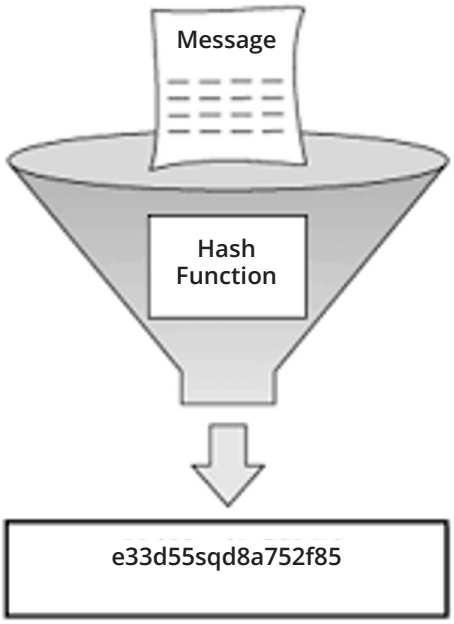
In ***Bitcoin***, every transaction is hashed before it is added to the *blockchain*. The hash acts as a signature for the transaction, verifying that the transaction is valid and has not been tampered with. If someone tries to change even a single letter in the transaction, the hash will be completely different, alerting others to the change.



The Role of Hashing in Providing Security

Hashing is essential to the security of the ***Bitcoin Network***. By using hashes to identify transactions, the network can detect any attempt to change or manipulate a transaction. This helps to prevent fraud and ensure that all transactions are recorded accurately on the *blockchain*.

A hash function is a type of **one-way function** that takes an **input** (referred to as the "message" or "data") and converts it into a numerical representation referred to as a "**hash**." The **output** hash is unique to the **input** data, so even a small change in the input data results in a completely different hash.



A **hash function** is like a secret code machine. It takes in a **message** and turns it into a code.

- The code always looks the same for the same message. If you change the message even a little, the code will be completely different. This helps computers remember things and check if anything has been changed.

Instantly generate a SHA256 hash of any string or input value. Hash functions are used as one-way methods.